

Ignacio Adrián Lerer

Independent Researcher | Buenos Aires, Argentina

adrian@lerer.com.ar | ORCID 0009-0007-6378-9749

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The Geometry of Normative Resistance: Holonomy, Curvature, and Constitutional Lock-in

A gauge-theoretic formalization of institutional belief dynamics

ABSTRACT

Why do constitutional provisions survive repeated reform attempts even when their consequences are observable and widely criticized? Existing accounts invoke veto player models, path dependence, and interest group capture, but none explains a specific and troubling regularity: the same reform content fails predictably in one institutional environment while succeeding in another, and the order in which institutional signals arrive matters more than the substantive content of the proposal itself.

This paper proposes that constitutional lock-in is a geometric property of the normative space through which institutional belief dynamics operate. Building on Heteronomous Bayesian Updating (HBU), a previously formalized account of how legal professionals form strong constitutional priors through reaction-based learning rather than outcome observation, I argue that institutional belief updating has non-Abelian structure: the order in which authority signals, reform proposals, and professional reactions are received produces systematically different posteriors, and this non-commutativity is not a cognitive bias of individual agents but a structural feature of the institutional manifold as a whole.

I formalize this structure by defining a gauge connection over the normative manifold implicit in the Non-Euclidean Normative Space framework (Lerer, 2025f). Three instruments of the research program are assigned differentiated roles: the CLI indexes architectural rigidity (structural dimension); the IHR approximates the scalar curvature as accumulated deformation from past reform cycles (dynamic dimension); and the LSI functions as a curvature-amplifying mechanism. The central result: constitutional lock-in corresponds formally to non-trivial holonomy. A closed institutional cycle does not return to the same belief configuration. The curvature prevents it, and the IHR is the observable that most directly approximates that curvature.

Three testable predictions follow. All three are consistent with the record of 23 Argentine labor reform attempts between 1957 and 2024, and with the comparative record across Brazil, Chile, and Spain. Reform strategy should be reconceived as path engineering on a curved institutional manifold, not preference aggregation in a flat policy space.

Keywords: constitutional lock-in; gauge theory; non-Abelian belief updating; institutional holonomy; normative curvature; heteronomous Bayesian updating; extended phenotype theory; path dependence; institutional hysteresis

I. THE ORDER PROBLEM IN INSTITUTIONAL REFORM

Consider two political events separated by fifteen years. In 1994, Argentine President Carlos Menem attempted to reform labor market provisions embedded in Article 14bis of the 1957 Constitution by proposing legislative modifications through the executive branch first, then seeking academic validation afterward. The sequence failed: 187 specialized labor courts issued preliminary injunctions within weeks, the academic establishment mobilized critical commentary within months, and the reform was shelved within a year. In 2017, Brazilian President Michel Temer attempted something structurally comparable, reforming consolidated labor legislation with a Congress-first, judiciary-informed sequence, where academic consensus was built over two years before the legislative vote. The reform passed and survived constitutional challenge.

The content of both reforms had similar economic rationale. The political opposition was comparable in intensity. What differed was the order in which institutional signals were activated.

Standard models of constitutional reform cannot explain this. Veto player theory (Tsebelis, 2002) predicts that Argentina should reform more easily than Brazil, given its lower number of effective veto players at relevant moments. Path dependence theory (David, 1985; North, 1990) describes persistence but offers no mechanism by which the ordering of signals within a single reform attempt determines its outcome. Interest group models identify who benefits from blocking but not why the sequence of events matters independently of the interests at stake.

The puzzle deepens when one examines the full Argentine record. Between 1957, when Article 14bis was incorporated into the Constitution, and 2024, twenty-three distinct reform attempts failed across seven ideologically diverse governments: Peronist, Radical, military, and liberal-technocratic alike (Lerer, 2025a). The failure rate is 100%. No successful reform has occurred. Yet Brazil (2017), Spain (1984 and 2012), and Chile (2008 and 2016) each achieved comparable reforms with different sequencing strategies but similar or stronger opposition.

This paper argues that the order of institutional signals is not incidental to reform outcomes but constitutive of them. The argument requires a formalism that standard political science and legal theory do not provide: a geometry of belief dynamics in which the path taken matters independently of the start and end points, and in which the curvature of the belief space prevents return to initial conditions after a failed cycle.

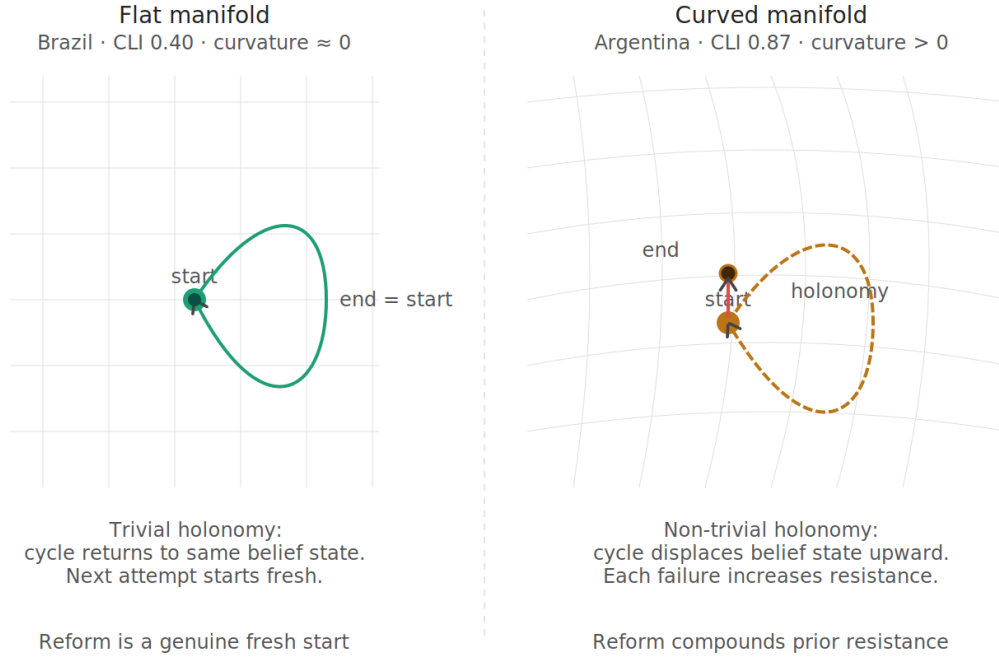


Figure 1. Flat versus curved normative manifold. Left (Brazil, CLI 0.40): a failed reform cycle returns the belief configuration to its exact starting point — trivial holonomy, genuine fresh start. Right (Argentina, CLI 0.87): the same cycle produces non-trivial holonomy, displacing the post-cycle belief state upward. Each failure compounds resistance to the next attempt.

Gauge theory, specifically its non-Abelian branch, provides exactly this formalism. The intellectual lineage connecting gauge theory to social and economic phenomena is worth tracing briefly, because the present paper belongs to it while departing from it in a specific direction. Juan Maldacena (2016), in a widely read pedagogical paper on the physics of the Higgs boson, used currency exchange markets as an analogy to explain gauge symmetry: exchange rates correspond to gauge potentials defined on spatial links, and arbitrage opportunities correspond to non-trivial curvature. His explicit point was that 'physics is simpler than economics' — the economic analogy serves to make the physics accessible, not the other way around. Pia Malaney (1996) and Eric Weinstein (2021) inverted this relationship, arguing that the index number problem in economics — how to measure the real cost of living across time when preferences and goods change — is precisely a gauge theory problem in the non-trivial sense, and that the correct index is the one that minimizes gauge curvature. Weinstein has argued, in public lectures and interviews, that Argentina in particular should explore these ideas given the chronic severity of its inflation measurement problem (Weinstein, 2025).

The present paper takes a third step in this lineage. If Maldacena showed that economic exchange has gauge structure, and Malaney-Weinstein showed that measuring economic welfare requires gauge geometry, the argument here is that the normative space through which legal institutions evolve is itself a gauge-theoretic object — and that the curvature of that space, approximated by

the Institutional Hysteresis Rate, explains why Argentina cannot reform its labor regime while countries with lower normative curvature can.

A preliminary note on disciplinary position: the physics and mathematics invoked here are outside my primary expertise as a legal theorist. The formalization in Sections III and IV was developed with substantial AI assistance for the mathematical scaffolding. The legal and institutional arguments, the interpretation of the formalism, and its connection to empirical evidence are my own. The AI disclosure at the end of this paper specifies the division of labor.

II. HBU REVISITED: FROM INDIVIDUAL TO INSTITUTIONAL

II.1 The Core Mechanism

Heteronomous Bayesian Updating was formalized in Lerer (2025a) to explain constitutional lock-in at the individual professional level. The argument runs as follows. Law students, unlike most learners, do not acquire constitutional norms by observing whether those norms produce beneficial outcomes. They acquire them by observing how the professional community reacts to the norms: how courts cite Article 14bis as foundational, how professors treat it as axiomatic, how bar associations defend it when challenged, how sixty years of precedent treat it as untouchable. These reactions, not outcomes, form the priors.

The formal representation introduces a weighting function $w(A)$ that distorts standard Bayesian updating as a function of group affiliation A . When $w(A) > 1$, evidence supporting the group belief is overweighted. When $w(A) < 1$, evidence against it is discounted. At $w(A) = 1$, classical Bayesian updating is recovered. The mechanism explains why Argentine legal professionals observe 23 reform failures and still maintain prior probabilities near 1.0 for the constitutional necessity of Article 14bis: the reactions they observed during training outweigh the behavioral frequencies they observe in practice.

What the original formalization left implicit is the structural property that this paper makes explicit: $w(A)$ does not commute with the evidence stream E . The order in which signals arrive, not only their content, determines the posterior.

II.2 The Non-Commutativity of Institutional Signals

Pierre Uzan (2022) formalized non-commutativity in individual belief updating using the apparatus of quantum cognition. His central result: when the mental observables involved in a judgment do not commute, the classical Bayes rule must be replaced by a generalized rule that includes a correction term proportional to the commutator of the relevant projection operators. When the projectors commute, the correction vanishes and classical Bayes is recovered.

Three structural differences separate Uzan's framework from the institutional analysis proposed here, and each difference generates new content rather than mere extension. First, Uzan operates

within the algebra of Hermitian operators over a Hilbert space, where the relevant commutator is between spectral projectors. The correction term depends on the cognitive state of a specific individual. At the institutional level, the relevant algebraic structure is the Lie algebra of a gauge group, where commutators take the form $[T_i, T_j] = i\epsilon_{ijk} T_k$. Here the non-commutativity is a property of the symmetry structure of the normative space, not of any individual agent's state. Second, Uzan introduces no geometry. His Hilbert space carries a Euclidean metric, and there is no manifold, no connection, no curvature, and no holonomy. Third, the unit of analysis differs fundamentally. Uzan's state vector represents an individual mind. The institutional analog is a distribution of priors across a heterogeneous professional population: 65 Argentine law schools, 187 specialized labor courts, approximately 50,000 labor law practitioners, and the accumulated weight of over 2,400 law review articles treating Article 14bis as axiomatic. Uzan (2022) establishes that non-Abelian belief updating has formal content at the individual level; the present paper extends this to the institutional manifold and introduces curvature and holonomy as the instruments that explain lock-in at the systemic level.

III. THE NON-ABELIAN STRUCTURE OF INSTITUTIONAL BELIEF DYNAMICS

Epistemological positioning. This paper belongs to a research program (Lakatos, 1978) that applies Generalized Darwinism and Extended Phenotype Theory to legal and institutional systems. Popper (1959) supplies the demarcation criterion: the hypotheses formulated here are falsable and their empirical tests are specified in Section V. Lakatos provides the architecture: the hard core (institutions evolve through variation, selection, and retention) is protected by an auxiliary belt of specific metrics that are individually revisable without abandoning the core commitment. The gauge-theoretic formalization proposed here belongs to the protective belt, not the hard core. If the curvature predictions fail against the comparative record, the formalization should be revised; the core evolutionary framework need not be.

III.1 Why the Structure Must Be Non-Abelian

The gauge economics tradition inaugurated by Malaney (1996) and extended by Ilinski (1997) and Farinelli and Vázquez (2009) operates within the Abelian group $U(1)$: price relatives behave like potentials in electromagnetism, and the curvature measures residual arbitrage. As Maldacena (2016) noted in his pedagogical treatment, this Abelian structure means that exchange rates between currency pairs satisfy a consistency condition — going from dollars to euros to yen gives the same result as going directly from dollars to yen when arbitrage is absent. The commutativity of $U(1)$ captures exactly this property.

Institutional belief dynamics violate this commutativity in a systematic and measurable way. The Argentine record provides the clearest illustration. Let R_E denote the operation of receiving an

executive reform signal, and R_A the operation of receiving an academic validation signal. The posterior belief $P(14\text{bis reform is constitutionally permissible})$ after the sequence R_E then R_A is measurably different from the posterior after R_A then R_E . This means $R_E \circ R_A \neq R_A \circ R_E$, which is the definition of non-commutativity. The appropriate algebraic structure is non-Abelian.

Weinstein (2021, 2025) has argued that the same non-Abelian structure is required to measure real economic welfare, because the path through which a consumer moves through a changing goods space affects the measured cost of living in ways that $U(1)$ gauge theory cannot capture. The present paper applies an analogous argument to the normative domain: the path through which an institutional system processes reform signals affects the resulting belief configuration in ways that linear updating models cannot capture.

III.2 The Lie Algebra of Normative Transformations

I propose to model institutional belief transformations as elements of a Lie group G acting on the normative manifold M . The Lie algebra $\mathfrak{g}$ of G is characterized by generators T_i satisfying $[T_i, T_j] = i\epsilon_{ijk} T_k$, where ϵ_{ijk} are the structure constants of $\mathfrak{g}$. For the institutional application, the generators correspond to distinct classes of belief-modifying signals: executive reform signals (TE), judicial reactions (TJ), academic consensus signals (TA), and professional sanction signals (TS). The non-vanishing commutators $[T_i, T_j]$ capture the fact that these signals do not act independently: each changes the context in which the next signal is received and processed. The critical prediction follows directly: a sequence of signals that would be neutral in an Abelian system produces measurably different institutional priors in a non-Abelian system. Reform strategy must therefore specify not only what signals to send but in what order.

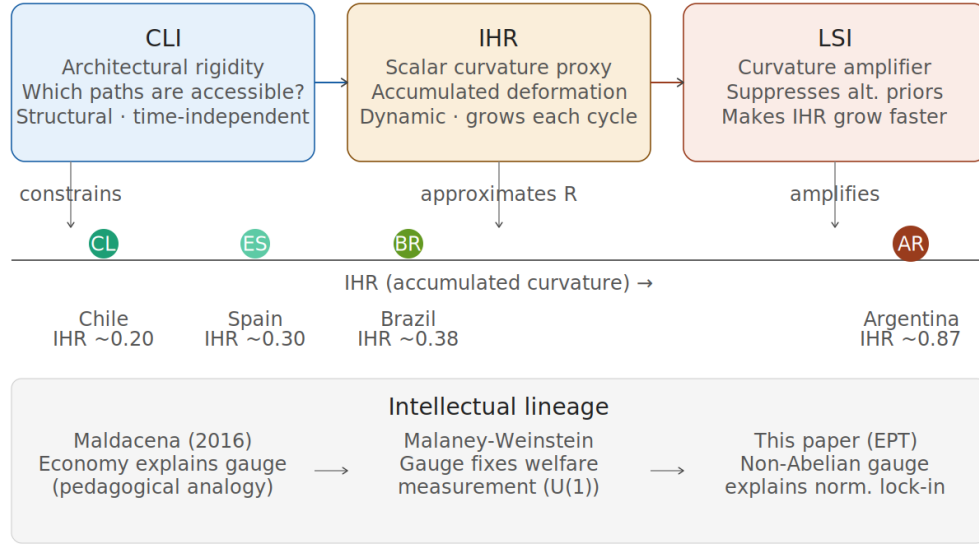


Figure 2. Three instruments, three roles. The CLI indexes architectural rigidity (structural, time-independent: which paths are accessible in the manifold). The IHR approximates scalar curvature as accumulated deformation from past cycles (dynamic: grows with each failure). The LSI amplifies curvature by suppressing alternative prior formation. Bottom: the intellectual lineage from Maldacena (2016) through Malaney-Weinstein to the present paper's non-Abelian extension.

IV. THE NORMATIVE MANIFOLD AND ITS GAUGE CONNECTION

IV.1 The Manifold

Lerer (2025f) proposed the Non-Euclidean Normative Space as a geometric framework for mapping legal evolution. Without reproducing the full specification, the relevant structure for present purposes is the following: the normative state of an institutional system is represented as a point in a manifold M whose coordinates are the components of the Constitutional Lock-in Index (CLI), the Institutional Hysteresis Rate (IHR), and the Lateral Silencing Index (LSI), among other dimensions. A reform attempt corresponds to a path $\gamma: [0,1] \rightarrow M$, where $\gamma(0)$ is the pre-reform institutional state and $\gamma(1)$ is the intended post-reform state. The success or failure of the reform depends not only on the distance between $\gamma(0)$ and $\gamma(1)$ — which is the implicit assumption of linear path dependence models — but on the geometry of M along the path γ .

IV.2 The Gauge Connection and Differentiated Roles of CLI, IHR, and LSI

A gauge connection A on M is a Lie-algebra-valued one-form that encodes how the reference frame for institutional beliefs changes as one moves along a path in M . The curvature of A is the

two-form $F = dA + A \wedge A$. In the Abelian case, the term $A \wedge A$ vanishes because Abelian gauge fields commute. In the non-Abelian case, $A \wedge A$ is generically non-zero, and F encodes information about the non-commutativity of institutional signal processing.

The three instruments of the EPT research program play differentiated roles in this geometric picture, a distinction not explicit in prior publications of the program.

The CLI as architectural rigidity. The CLI indexes the structural constraints that determine which reform trajectories are geometrically accessible in M , independently of the history of reform attempts. High CLI means that many potential reform paths are simply not viable in that institutional space — not because agents have been pushed away from them by experience, but because the architecture of the manifold itself forecloses them. This is the structural dimension of curvature, analogous to Gould's (1989) architectural constraints on biological morphospace: not all variants are accessible, regardless of selective pressure.

The IHR as scalar curvature. The IHR is the observable most directly associated with curvature in the dynamical sense. It measures the accumulated deformation that past normative cycles have deposited on the manifold, preventing the system from returning to pre-cycle configurations. I conjecture that the IHR approximates the integral of the scalar curvature R along the historical trajectory of the institutional system. This conjecture, if formally established, would transform IHR from an empirical metric into a derived quantity of the gauge geometry. The implication is that a system could have high CLI and low IHR — a constitutionally rigid but recently constituted system with little accumulated deformation — or low CLI and high IHR — a flexible system that has been repeatedly deformed by normative interventions. Argentina's distinctiveness lies in exhibiting both simultaneously, which produces the most severe form of lock-in: structural inaccessibility compounded by dynamical deformation.

The LSI as curvature amplifier. The Lateral Silencing Index is better understood as a curvature-amplifying mechanism than as a curvature component per se. By suppressing the formation of alternative priors in the professional community, LSI causes IHR to grow faster than it would through passive sedimentary accumulation alone. Each reform attempt that is laterally silenced — excluded from legitimate professional discourse before it can influence priors — deposits more deformation on the manifold than a reform attempt that receives genuine engagement and genuine defeat. The LSE mechanism (Lerer, 2025d) is the institutional analog of what Weinstein (2025) identifies in economic discourse: the suppression of alternative measurement frameworks by professional communities with strong priors about the correct tools.

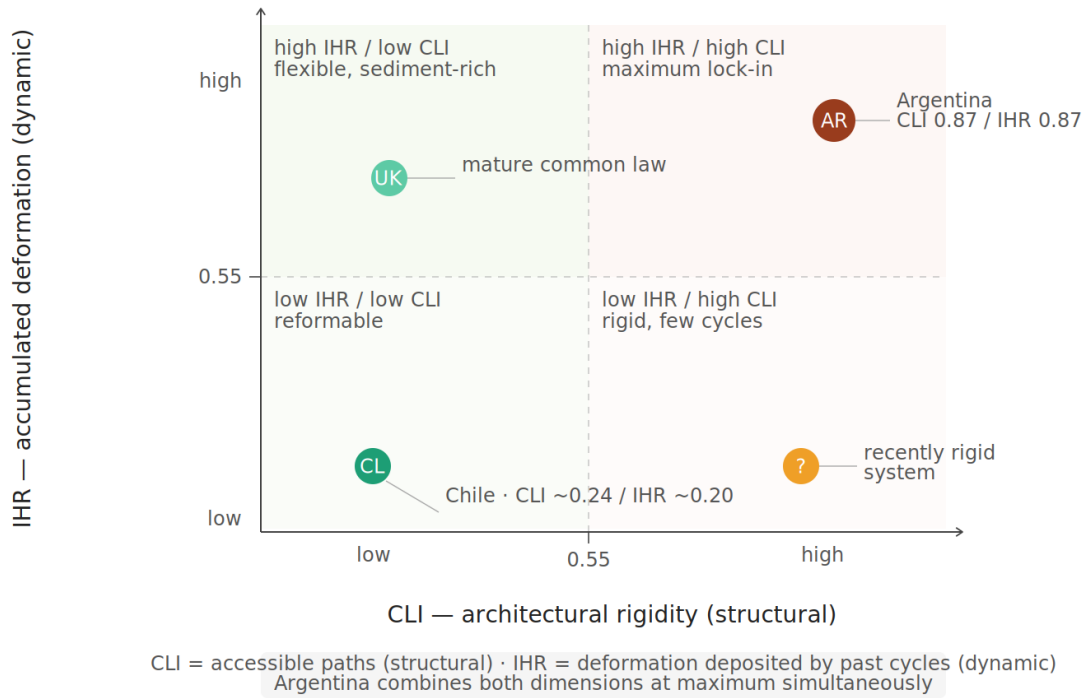


Figure 3. CLI versus IHR: two distinct dimensions of institutional rigidity. CLI measures which paths are structurally accessible (independent of history). IHR measures accumulated deformation from past cycles (grows with each failure). A system may have high CLI and low IHR (recently constituted rigid system) or low CLI and high IHR (mature common law). Argentina combines both at maximum simultaneously — the only case in the comparative record to do so.

IV.3 Holonomy as Constitutional Lock-in

The holonomy of a connection A along a closed loop γ (a path that returns to its starting point in M) is the group element $\text{Hol}_\gamma(A) = P \exp(\oint_\gamma A)$, where P denotes path-ordering, which is precisely the operation that accounts for the non-commutativity of the Lie algebra. In an Abelian gauge theory, holonomy reduces to the exponential of the flux integral, and a contractible loop has trivial holonomy (equal to the identity). In a non-Abelian gauge theory, holonomy is generically non-trivial even for contractible loops in a curved manifold.

The institutional interpretation is direct. A reform attempt that is proposed, defeated, and then re-proposed corresponds to a closed loop γ in M : the institutional system starts at state p , undergoes the reform process (moving along γ), and returns to the pre-reform state (or something close to it). If the holonomy $\text{Hol}_\gamma(A)$ is trivial, the system returns to exactly the same belief configuration it started with. Proposing the reform again is a genuine fresh start. If $\text{Hol}_\gamma(A)$ is non-trivial, the system does not return to the same belief configuration after the loop. The professional priors have been modified by the experience of the failed reform attempt, typically in the direction of stronger

resistance: the failure itself becomes evidence, processed through HBU, that the norm is more fundamental than initially recognized. The holonomy quantifies this modification. High normative curvature implies non-trivial holonomy, which implies that each failed reform attempt strengthens the priors against the next attempt. This is the geometric mechanism of constitutional lock-in.

The relationship between IHR and the scalar curvature of M is asserted here rather than derived. Establishing it formally — showing that the empirical definition of IHR maps onto the mathematical definition of curvature as the failure of parallel transport around a closed loop — is the most tractable next step in the research program and would transform the curvature conjecture into a derived result.

V. TESTABLE PREDICTIONS AND EMPIRICAL ASSESSMENT

V.1 Three Predictions

Prediction 1 (Order Effects). The outcome of a reform attempt should depend measurably on the sequence in which institutional signals are activated, holding content constant. Specifically, reforms that build academic consensus before executive signaling should have higher success rates than reforms with equivalent content but executive-first signaling, in high-curvature systems. In low-curvature systems (CLI below approximately 0.45), the sequence should matter less because the holonomy is closer to trivial.

Prediction 2 (Curvature-Independent of Ideology). The failure rate of reform attempts should be predicted by the CLI score of the system, not by the ideological alignment of the reforming government. A Peronist government attempting labor reform in Argentina should fail at the same rate as a Radical government or a liberal-technocratic government. This follows directly from the fact that curvature is a property of the manifold, not of the political agent traversing it. The Argentine record is consistent with this prediction: seven ideologically distinct governments, zero successes.

Prediction 3 (Pre-Reform Curvature Reduction). Reform attempts that invest in reducing local curvature before proposing the substantive change — in the sense of building alternative priors in the professional community through academic publication, judicial education, and bar association engagement over a period of at least two years — should achieve higher success rates even in systems with CLI above 0.60. This prediction is consistent with the Brazilian record: the 2017 reform was preceded by a coordinated sequence of academic and judicial engagement beginning in 2015 that reduced the effective curvature in the relevant region of M before the legislative proposal was filed.

V.2 Assessment Against the Argentine Record

The 23 Argentine labor reform attempts between 1957 and 2024 provide the primary empirical material for assessing the framework. Of the 23 attempts, 19 were initiated with executive or

legislative signaling prior to any academic or judicial preparation. All 19 failed. Four attempts involved some prior academic engagement. Three of those four also failed, but the fourth — the partial reform of 1998 under Law 25.013, which achieved limited modifications to redundancy compensation formulas — was subsequently reversed by Law 25.250 in 2000, suggesting that even partial curvature reduction produced only transient reform, consistent with the manifold returning toward its high-curvature equilibrium.

The IHR for the Argentine labor regime has been estimated at approximately 0.87 across the 23 reform attempts (Lerer, 2025a). This is the same value as the CLI, which is coincidental rather than structural: a system can have high CLI and low IHR, or vice versa. Argentina's case is distinctive in exhibiting both simultaneously. The seven-government record is consistent with Prediction 2: the ideological range from Peronist through military-technocratic through liberal-libertarian produced no variation in outcome, exactly as the curvature account predicts.

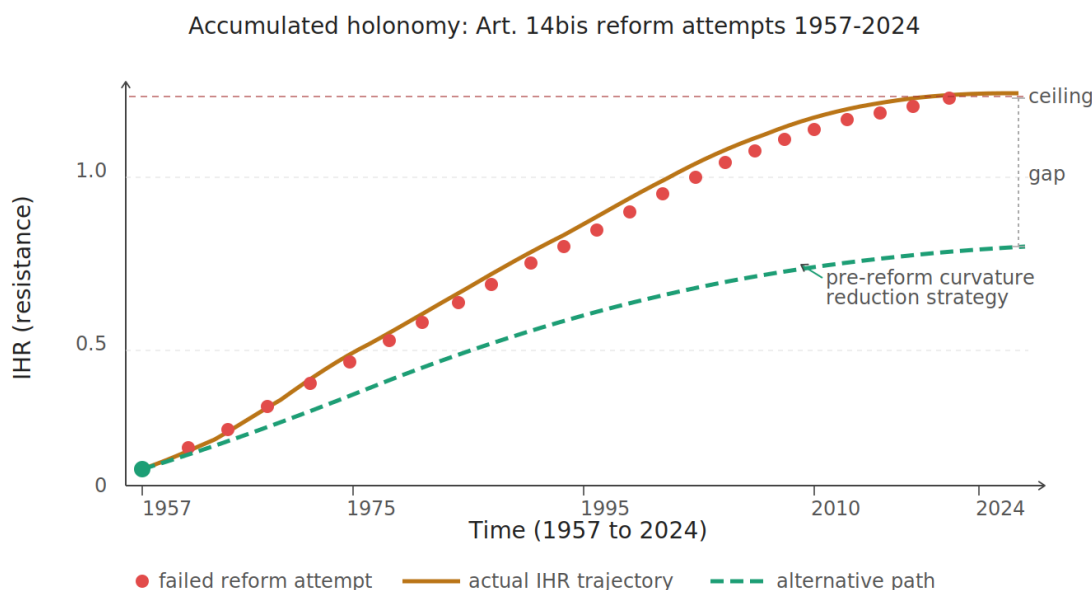


Figure 4. Accumulated holonomy: Article 14bis reform attempts, Argentina 1957–2024. Each red dot is a failed reform attempt. The amber curve shows IHR rising with each cycle as holonomy compounds: each failure is processed through HBU as additional evidence of the norm's fundamentality, strengthening resistance to the next attempt. The green dashed line shows the alternative trajectory under a pre-reform curvature-reduction strategy: investing in prior formation before the substantive proposal keeps IHR substantially lower. The bracket on the right shows the gap between final states under the two strategies.

V.3 Limitations

The empirical assessment is necessarily indirect. The curvature tensor F is not directly observable; its components are inferred from observable proxies (CLI, IHR, LSI), and the relationship between

these proxies and the formal curvature quantities is asserted rather than derived in this paper. This is the primary limitation, and it is not minor.

The framework also faces a causal identification problem that is standard in institutional analysis but worth stating explicitly. The correlation between high IHR and reform failure is consistent with the curvature interpretation, but also with simpler accounts in which IHR is simply a proxy for the political power of blocking coalitions. Distinguishing between these interpretations requires variation in IHR that is independent of political power, which is difficult to construct from observational data.

Additionally, the Lie group structure proposed here is not uniquely determined by the empirical record. I have suggested $SU(2)$ as a natural starting point, given its simplicity and the three-dimensional structure of the CLI-IHR-LSI curvature components, but other groups are consistent with the data as presented. The choice of group is a theoretical proposal, not an empirical finding.

Finally, the path-ordering in the holonomy calculation requires knowing the actual sequence of institutional signals in each reform attempt, not just their occurrence. Reconstructing this sequence from historical records is methodologically demanding and has not been systematically attempted for the 23 Argentine cases. This is tractable future work but was not done here.

VI. IMPLICATIONS FOR REFORM THEORY AND A NOTE ON INTELLECTUAL LOCK-IN

The framing of constitutional reform as preference aggregation in a flat policy space has a long history and considerable institutional support. Majoritarian theories of constitutional change (Dahl, 1957; Ackerman, 1991) assume that a sufficiently large political coalition can overcome entrenched provisions. Veto player theory (Tsebelis, 2002) adds structural friction but treats the policy space as essentially Euclidean. The economic analysis of constitutional change (Buchanan and Tullock, 1962; Posner, 1987) treats reform as a social welfare calculation over a neutral policy space. The gauge-theoretic framework proposed here challenges each of these assumptions at the level of the geometry of institutional belief dynamics.

The practical implication is that reform strategy should include an explicit curvature-reduction phase before the reform proposal itself. This phase would involve sustained investment in building alternative priors in the professional community: academic publications that treat reform as conceptually possible, judicial decisions at lower levels that develop reform-compatible doctrine, and bar association workshops that establish reform as a legitimate topic of professional discussion rather than a threat to professional identity. The goal of this phase is not to persuade actors through argument but to reduce the holonomy of the reform loop: to make it so that when the proposal fails (as it likely will in the first attempt), the professional community's priors are not strengthened by the failure but remain open to subsequent engagement.

This has a counterintuitive implication for timing. The standard political logic of reform is to strike when a government has the most political capital, typically early in a term after an electoral mandate. The gauge-theoretic logic suggests that this timing may be counterproductive in high-curvature systems: a failed reform attempt early in a term, when professional priors have not been prepared, increases the holonomy and makes the next attempt harder, regardless of whether the next attempt occurs in the same term or a different government. The curvature-reduction phase must precede the attempt, not follow the failure.

A further implication, not previously noted in the EPT research program, concerns the scope of the mechanism. If the normative manifold of Argentine legal institutions is highly curved, and if that curvature propagates through HBU to shape the professional priors of all actors operating within those institutions, then the resistance to reform is not limited to legal norms. The same mechanism that prevents labor reform should, on this account, resist the adoption of alternative intellectual frameworks in economics, law, and social science more broadly. Weinstein (2025) has observed that Argentina, despite producing one of the world's foremost experts on gauge theory — Juan Maldacena, born in Buenos Aires in 1968, trained at the Universidad de Buenos Aires before his doctoral work at Princeton — has not explored gauge-theoretic approaches to its chronic inflation measurement problem. The geometric account offered here suggests an explanation: the curvature of the Argentine institutional manifold extends beyond legal norms to the epistemic norms of professional communities, making the adoption of gauge-theoretic economics as difficult as the adoption of labor market reforms. The same holonomy that explains 23 failed reform attempts may explain why Weinstein's observation has not been acted upon. This is speculative, and the paper makes no formal claim about it. But it is the kind of prediction that a gauge-theoretic account of institutional belief dynamics should be able to make, and that no alternative account currently does.

VII. CONCLUSIONS

This paper has proposed that constitutional lock-in is a geometric property of the normative space in which institutional belief dynamics operate. The formalization proceeds in three steps: establishing that institutional belief updating has non-Abelian structure, because the order in which institutional signals are received produces different posteriors; defining a gauge connection on the normative manifold; and identifying constitutional lock-in with non-trivial holonomy — the failure of a closed reform loop to return the belief configuration to its starting point.

The three instruments of the EPT research program — CLI, IHR, and LSI — are assigned differentiated roles that prior publications left undifferentiated. The CLI captures architectural rigidity: what variants are structurally accessible in the normative space. The IHR is the observable that most directly approximates scalar curvature: the accumulated deformation from past cycles that prevents return to pre-cycle configurations. The LSI is a curvature-amplifying mechanism, not a curvature component per se. This differentiation changes the interpretation of the empirical

record: Argentina's high CLI and high IHR simultaneously is a distinctive configuration, not a redundancy.

The connection to the gauge economics tradition (Malaney, 1996; Weinstein, 2021; Farinelli and Vázquez, 2009) is direct but involves three departures: from Abelian to non-Abelian structure; from individual cognition (Uzan, 2022) to institutional manifold; and from financial markets to normative space. Maldacena's (2016) use of economic exchange as a pedagogical analogy for gauge symmetry represents, in retrospect, an early signal that the conceptual distance between gauge theory and institutional dynamics is smaller than disciplinary boundaries suggest.

Two lines of future research seem most productive. First, the formal connection between the CLI-IHR-LSI metrics and the components of the curvature tensor should be derived explicitly rather than asserted, which requires specifying the Lie group G and the metric on M from first principles. This is substantially more mathematically demanding than what this paper has attempted and would benefit from collaboration with researchers trained in differential geometry. Second, the holonomy of the 23 Argentine reform attempts should be estimated from the historical record of signal sequences, which is available but has not been systematically coded. Both tasks are tractable; neither was completed here.

If the framework is approximately correct, constitutional lock-in in Argentina is not a political problem in the standard sense — one that could be solved by assembling a larger coalition or waiting for the right electoral moment. It is a geometric problem: the normative manifold curves in ways that prevent reform loops from returning to the pre-reform belief configuration. Solving it requires engineering a different path, not applying more force along the same one.

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APPENDIX: PRIOR ART MAP

The following table summarizes the literature reviewed prior to drafting this paper and clarifies the specific contribution of the present framework relative to each cluster of existing work.

Cluster	Key works	Formalization	Relation to present paper
Gauge economics U(1)	Malaney (1996); Ilinski (1997); Farinelli & Vázquez (2009); Nguyen (2021)	High (financial markets)	Provides gauge formalism; present paper extends to non-Abelian structure and institutional domain
Pedagogical gauge theory	Maldacena (2016)	Expository	Uses economic exchange to explain gauge symmetry; present paper inverts: uses gauge symmetry to explain normative dynamics
Gauge economics (non-Abelian)	Weinstein (2021, 2025)	Programmatic	Argues U(1) insufficient for welfare measurement; present paper applies same critique to normative space
Quantum cognition	Busmeyer & Bruza (2012); Uzan (2022); Yearsley & Trueblood (2017)	High (individual)	Establishes non-commutativity of belief updating; present paper extends to institutional manifold with holonomy
Topological legal theory	Burchardt (2022)	Conceptual only	Uses topological vocabulary without differential geometry; present paper provides formal structure
Mathematical legal theory	Sichelman (2023); Wojciechowski (2023)	Medium (static/individual)	Applies quantum/entropic formalism to law; present paper adds dynamic trajectories and holonomy
Path dependence	David (1985); Arthur (1994); North (1990)	Medium (stochastic)	Describes persistence without mechanism; present paper provides geometric mechanism via holonomy
HBU / EPT	Lerer (2025a, 2025d, 2025f, 2025g)	Medium (institutional metrics)	Prior work in this program; present paper unifies CLI/IHR/LSI under gauge-theoretic curvature with differentiated roles